AI-Augmented Security Operations Center for Real-Time Network Threat Detection Using Suricata and Splunk

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Abstract

Modern organizations face increasing cybersecurity threats targeting network infrastructures. Security Operations Centers (SOCs) are responsible for monitoring and responding to these threats. However, traditional SOC systems rely heavily on signature-based detection techniques that are limited in detecting unknown or evolving attacks.

This research presents an AI-augmented SOC architecture integrating Suricata Intrusion Detection System (IDS), machine learning-based anomaly detection using the Isolation Forest algorithm, and Splunk Security Information and Event Management (SIEM) for visualization and alerting. The system collects network traffic logs, extracts relevant features, trains a machine learning model to detect anomalous behavior, and visualizes security insights through a SOC dashboard.

Experimental evaluation demonstrates that the proposed system can automatically identify suspicious network activities and generate automated alerts to assist SOC analysts in responding to potential threats.

Keywords: SOC, Intrusion Detection System, Suricata, Machine Learning, Isolation Forest, Splunk SIEM, Cybersecurity

1 Introduction

Cybersecurity has become one of the most critical challenges for modern organizations. As enterprise networks continue to grow in size, complexity, and exposure, detecting malicious activities in real time becomes increasingly difficult. Security Operations Centers (SOCs) are responsible for continuously monitoring, analyzing, and responding to suspicious events across network and system environments.

Traditional intrusion detection systems mainly rely on signature-based methods. These approaches compare observed traffic against known attack signatures and are effective in recognizing previously identified threats. However, they are significantly less effective against zero-day attacks, previously unseen attack patterns, and subtle anomalous behaviors that do not exactly match predefined rules.

Artificial intelligence and machine learning offer a promising enhancement to modern SOC operations. Instead of depending only on known signatures, machine learning

models can identify deviations from normal behavior and flag suspicious events for further investigation. This capability is especially important in environments where large volumes of logs make manual inspection inefficient.

This project proposes an AI-augmented SOC platform that integrates Suricata IDS, a Python-based feature extraction pipeline, an Isolation Forest anomaly detection model, and Splunk SIEM for visualization and automated alerting. The goal is to create a practical end-to-end detection workflow capable of identifying suspicious network behavior and supporting SOC analysts with actionable insights.

2 Research Contributions

The main contributions of this research are summarized as follows:

- Design of an AI-augmented SOC architecture integrating Suricata IDS, machine learning, and SIEM visualization.
- Development of a Python-based feature extraction pipeline for processing Suricata network logs.
- Implementation of an Isolation Forest anomaly detection model for identifying suspicious network behavior.
- Integration of anomaly detection results into a Splunk SOC dashboard for real-time monitoring.
- Demonstration of an end-to-end detection workflow for cybersecurity operations.

3 Related Work

Intrusion detection has been extensively studied in cybersecurity research. Signature-based IDS solutions such as Snort and Suricata are widely adopted because of their efficiency and mature rule ecosystems. These systems are effective for known attacks but have limitations in detecting novel or low-signal malicious activities.

Recent work in cybersecurity has explored anomaly detection using machine learning models such as Support Vector Machines, Decision Trees, Random Forests, Autoencoders, and Isolation Forests. Among these, Isolation Forest is particularly attractive for network security use cases because it is unsupervised, computationally efficient, and suitable for datasets where labeled attack samples are limited.

Researchers have also investigated the integration of machine learning outputs into SIEM environments. Such integration enhances security visibility by correlating AI-driven anomaly detection with operational dashboards, threat monitoring panels, and automated alert mechanisms. This project follows that direction by combining IDS-generated network evidence with machine learning-based analysis and SIEM visualization.

Recent research has explored the use of machine learning techniques for network intrusion detection. Approaches using Support Vector Machines, Random Forests, and deep learning models such as Autoencoders have shown promising results in identifying anomalous network behavior. Datasets such as KDD Cup 1999 and CICIDS2017 are commonly used for evaluating IDS models.

Several studies have also investigated the integration of anomaly detection with Security Information and Event Management (SIEM) platforms. Combining machine learning detection with SIEM dashboards improves threat visibility and enables SOC analysts to prioritize suspicious activities efficiently.

4 System Architecture

The proposed AI-augmented SOC architecture consists of four major components:

- Suricata Intrusion Detection System
- Feature Extraction Engine
- Machine Learning Anomaly Detection Module
- Splunk SIEM Dashboard and Alerting System

The workflow begins with Suricata IDS monitoring network traffic and generating structured logs in JSON format through the *eve.json* file. These logs are transferred to the AI processing environment, where a Python script extracts relevant traffic and alert features. The processed dataset is then used to train and execute an Isolation Forest anomaly detection model. Finally, anomaly outputs are ingested into Splunk for analysis, dashboard visualization, and alert generation.

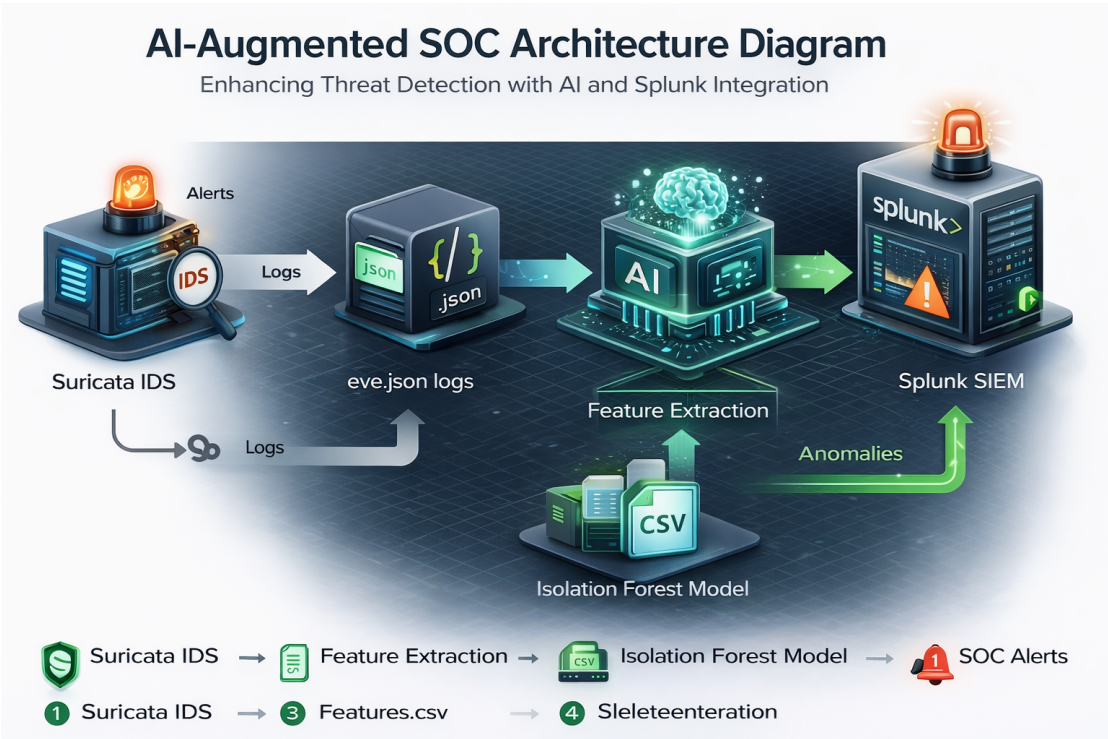


Figure 1: AI-Augmented SOC Architecture

The architecture demonstrates how traditional rule-based detection and machine learning-based anomaly detection can complement each other within the same SOC workflow. Suricata provides packet-level visibility and alert metadata, while the AI engine detects suspicious patterns that may not be obvious from static signature matching alone.

5 Data Collection

Network traffic data was collected using Suricata IDS deployed within a virtualized environment. Suricata inspects network packets in real time and records detailed flow and alert information in the *eve.json* log file. Each log entry contains metadata describing traffic characteristics and alert context.

Important attributes captured from the logs include:

- Source IP address
- Destination IP address
- Source port
- Destination port
- Protocol type
- Packet counters
- Byte transfer statistics
- Alert signatures and severity

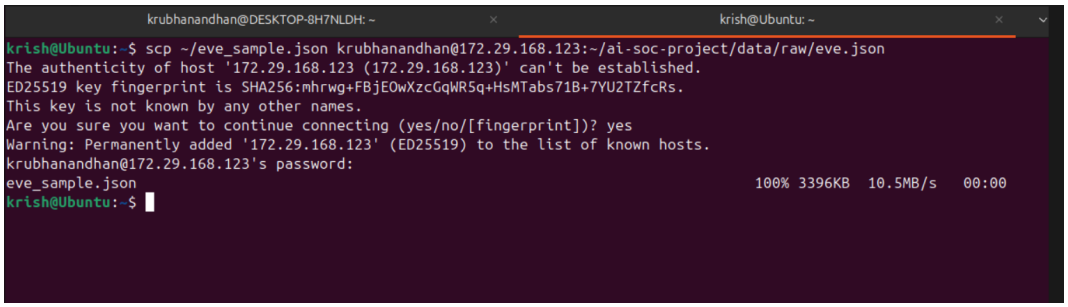
Table 1: Extracted Network Traffic Features

Feature	Description
src_ip	Source IP address
dest_ip	Destination IP address
src_port	Source port number
dest_port	Destination port number
flow_pkts_toserver	Packets sent to server
flow_pkts_toclient	Packets sent to client
flow_bytes_toserver	Bytes sent to server
flow_bytes_toclient	Bytes sent to client
alert_severity	Suricata alert severity level

For this project, a subset of approximately 5000 network events was selected from the Suricata output and transferred securely to the AI processing environment. This dataset was sufficient to demonstrate the end-to-end pipeline, including preprocessing, anomaly detection, and SIEM-based monitoring.

```
krish@Ubuntu:~$ ls -lh /var/log/suricata/eve.json
-rw-r--r-- 1 krish krish 2.0G Mar  8 12:13 /var/log/suricata/eve.json
krish@Ubuntu:~$ sudo tail -n 10 /var/log/suricata/eve.json
{"timestamp":"2026-03-08T12:14:22.345236+0000","flow_id":2180578179329873,"in_iface":"enp0s3","event_type":"alert","src_
ip":"172.29.168.123","src_port":9997,"dest_ip":"10.0.2.15","dest_port":49646,"proto":"TCP","pkt_src":"wire/pcap","alert"
:{"action":"allowed","gid":1,"signature_id":2210029,"rev":2,"signature":"SURICATA STREAM ESTABLISHED invalid ack","categ
ory":"Generic Protocol Command Decode","severity":3},"app_proto":"failed","direction":"to_client","flow":{"pkts_toserver
":71897,"pkts_toclient":71033,"bytes_toserver":108633843,"bytes_toclient":4262162,"start":"2026-03-08T12:07:11.048953+00
00","src_ip":"10.0.2.15","dest_ip":"172.29.168.123","src_port":49646,"dest_port":9997}}
{"timestamp":"2026-03-08T12:14:22.345236+0000","flow_id":2180578179329873,"in_iface":"enp0s3","event_type":"alert","src_
ip":"172.29.168.123","src_port":9997,"dest_ip":"10.0.2.15","dest_port":49646,"proto":"TCP","pkt_src":"wire/pcap","alert"
:{"action":"allowed","gid":1,"signature_id":2210045,"rev":2,"signature":"SURICATA STREAM Packet with invalid ack","categ
ory":"Generic Protocol Command Decode","severity":3},"app_proto":"failed","direction":"to_client","flow":{"pkts_toserver
":71897,"pkts_toclient":71033,"bytes_toserver":108633843,"bytes_toclient":4262162,"start":"2026-03-08T12:07:11.048953+00
00","src_ip":"10.0.2.15","dest_ip":"172.29.168.123","src_port":49646,"dest_port":9997}}
{"timestamp":"2026-03-08T12:14:22.345587+0000","flow_id":2180578179329873,"in_iface":"enp0s3","event_type":"alert","src_
ip":"172.29.168.123","src_port":9997,"dest_ip":"10.0.2.15","dest_port":49646,"proto":"TCP","pkt_src":"wire/pcap","alert"
:{"action":"allowed","gid":1,"signature_id":2210029,"rev":2,"signature":"SURICATA STREAM ESTABLISHED invalid ack","categ
ory":"Generic Protocol Command Decode","severity":3},"app_proto":"failed","direction":"to_client","flow":{"pkts_toserver
":71897,"pkts_toclient":71034,"bytes_toserver":108633843,"bytes_toclient":4262222,"start":"2026-03-08T12:07:11.048953+00
```

Figure 2: Suricata IDS Generating Alert Logs in *eve.json*



```
krubhanandhan@DESKTOP-8H7NLDH: ~                    krish@Ubuntu: ~
krish@Ubuntu:~$ scp ~/eve_sample.json krubhanandhan@172.29.168.123:~/ai-soc-project/data/raw/eve.json
The authenticity of host '172.29.168.123 (172.29.168.123)' can't be established.
ED25519 key fingerprint is SHA256:mhrwg+FBjEOwXzcGqWR5q+HsMTabs71B+7YU2TZfcRs.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.29.168.123' (ED25519) to the list of known hosts.
krubhanandhan@172.29.168.123's password:
eve_sample.json                                                    100% 3396KB  10.5MB/s   00:00
krish@Ubuntu:~$
```

Figure 3: Secure Transfer of Suricata Logs to the AI Processing Environment

6 Feature Extraction

Raw Suricata logs contain rich but semi-structured information. To make the data suitable for machine learning, a Python-based preprocessing pipeline was implemented. The script reads each JSON log entry, parses important fields, handles missing values, and converts the data into a structured CSV format.

The selected features for the anomaly detection model included:

- Source port
- Destination port
- Flow packets to server
- Flow packets to client
- Flow bytes to server
- Flow bytes to client
- Alert severity

These features were chosen because they provide a compact yet informative representation of communication behavior and alert context. After extraction, the processed dataset was saved as *features.csv*, which served as the training and inference input for the machine learning model.

```
(venv) krubhanandhan@DESKTOP-8H7NLDH:~/ai-soc-project$ python ai/extract_features.py
Extracted 5000 records to /home/krubhanandhan/ai-soc-project/data/processed/features.csv
                        timestamp event_type  ...                                alert_signature                    alert_category
0  2026-03-08T12:26:45.698534+0000      alert  ...           SURICATA STREAM ESTABLISHED invalid ack  Generic Protocol Command Decode
1  2026-03-08T12:26:45.698534+0000      alert  ...           SURICATA STREAM Packet with invalid ack  Generic Protocol Command Decode
2  2026-03-08T12:26:45.698803+0000      alert  ...           SURICATA STREAM ESTABLISHED invalid ack  Generic Protocol Command Decode
3  2026-03-08T12:26:45.698803+0000      alert  ...           SURICATA STREAM Packet with invalid ack  Generic Protocol Command Decode
4  2026-03-08T12:26:45.939647+0000      alert  ...  SURICATA STREAM ESTABLISHED packet out of window  Generic Protocol Command Decode

[5 rows x 15 columns]
(venv) krubhanandhan@DESKTOP-8H7NLDH:~/ai-soc-project$
```

Figure 4: Feature Extraction Pipeline Converting Raw Logs into Structured CSV Data

7 Machine Learning Model

The anomaly detection component of the proposed system is based on the Isolation Forest algorithm. Isolation Forest is an unsupervised learning method specifically designed to detect anomalies by isolating unusual observations through random partitioning.

The main advantage of Isolation Forest is that anomalies are typically easier to isolate than normal points. This makes the model suitable for cybersecurity use cases where malicious traffic often behaves differently from common baseline traffic. Another benefit is that the algorithm does not require extensive labeled data, which is often difficult to obtain in real-world SOC environments.

The Isolation Forest model was configured with the following parameters:

- Number of estimators: 100
- Contamination rate: 0.05
- Random state: 42
- Feature scaling: StandardScaler normalization

The model was trained using the extracted features from the Suricata dataset. Once the model was fitted, it was serialized and stored for later reuse in anomaly inference.

```
(venv) krubhanandhan@DESKTOP-8H7NLDH:~/ai-soc-project$ python ai/train_model.py
Model trained successfully
Model saved to models/isolation_forest_model.pkl
(venv) krubhanandhan@DESKTOP-8H7NLDH:~/ai-soc-project$
```

Figure 5: Training the Isolation Forest Model

8 Experimental Results

The implemented pipeline processed approximately 5000 network events extracted from Suricata logs. After model execution, the Isolation Forest algorithm identified 250 anomalous events. This corresponds to roughly 5% of the processed dataset, which is consistent with the configured contamination level and the suspicious behavior observed in the captured traffic.

Table 2: Anomaly Detection Results

Metric	Value
Total Events Processed	5000
Detected Anomalies	250
Anomaly Ratio	5%

The detected anomalies included events associated with unusual packet behavior, suspicious connection characteristics, and Suricata-generated alerts related to stream irregularities and protocol decoding issues. These results demonstrate that the AI layer was able to isolate potentially risky events from the larger traffic dataset.

```
(venv) krubhanandhan@DESKTOP-8H7NLDH:~/ai-soc-project$ python ai/detect_anomalies.py
Total events: 5000
Anomalies detected: 250
Results saved to output/anomalies.csv
(venv) krubhanandhan@DESKTOP-8H7NLDH:~/ai-soc-project$ |
```

Figure 6: AI-Based Anomaly Detection Results Showing Flagged Suspicious Events

The anomaly detection output was exported to *anomalies.csv* and prepared for SIEM ingestion. This output provided a filtered and analyst-friendly dataset containing only suspicious events, reducing the workload for manual triage.

9 SOC Dashboard and Alerting

To provide operational visibility, the anomaly results were ingested into Splunk SIEM and visualized through a dedicated SOC dashboard. The dashboard was designed to support a SOC analyst workflow by presenting the most relevant indicators in a clear and actionable form.

The dashboard included the following monitoring panels:

- Total number of AI-detected anomalies
- Top suspicious source IP addresses
- Most frequent alert signatures
- Most targeted destination ports
- Protocol distribution
- Anomaly trend over time

These visualizations allow analysts to understand the scale, origin, and nature of suspicious activities. By correlating model output with Suricata alert metadata, the dashboard improves situational awareness and accelerates investigation.

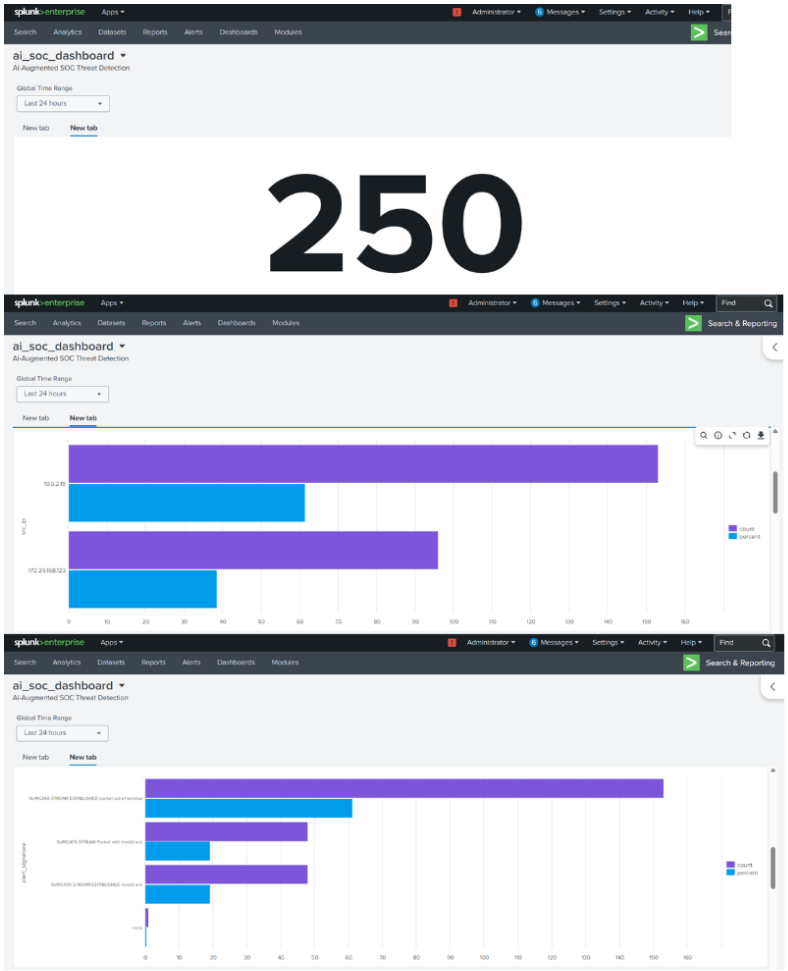


Figure 7: Splunk SOC dashboard presenting AI-detected anomalies across multiple monitoring panels including anomaly counts, suspicious source IP addresses, alert signature distribution, destination ports, protocol types, and anomaly trends over time.

In addition to visualization, an automated Splunk alert was configured to trigger when anomaly counts exceeded a defined threshold within a recent time window. This alerting mechanism demonstrates how AI-assisted detection can be operationalized into SOC response workflows.

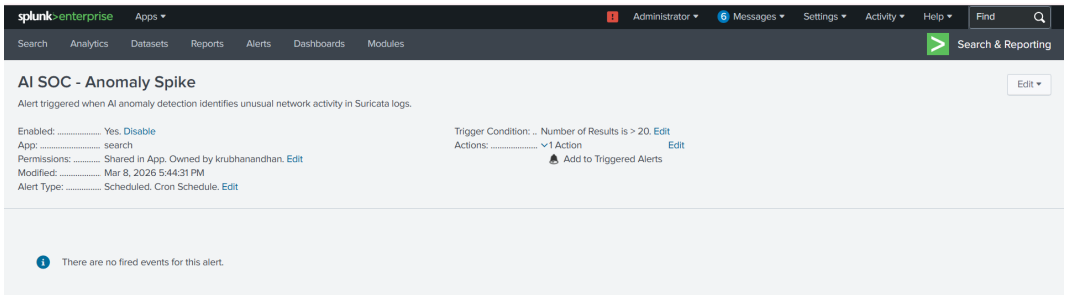


Figure 8: Automated Splunk Alert Triggered by Elevated Anomaly Activity

10 Threat Analysis

The anomalies detected by the machine learning model correspond to unusual network communication patterns observed in the Suricata logs. Many of the flagged events were associated with TCP stream irregularities, invalid acknowledgement packets, and protocol decoding issues.

Such anomalies may indicate suspicious activities such as network scanning, malformed packet injection, misconfigured network services, or potential reconnaissance behavior. By isolating these events from the larger traffic dataset, the proposed system assists SOC analysts in focusing on the most relevant security incidents.

11 Discussion

The project demonstrates the practical value of integrating machine learning into SOC workflows. Signature-based IDS tools remain essential for identifying known attack patterns and providing detailed packet-level context. However, when combined with an anomaly detection engine, the detection pipeline becomes more adaptive and capable of highlighting unusual behaviors that warrant analyst attention.

A major strength of this implementation is its modular design. Each stage of the workflow is clearly separated: detection, log transfer, preprocessing, model training, anomaly inference, SIEM ingestion, dashboarding, and alerting. This makes the architecture extensible and suitable for future enhancements.

Some limitations remain. The current implementation used a subset of 5000 events rather than a continuous real-time stream, and the anomaly detection model was based on a limited set of manually selected features. Future versions could incorporate larger datasets, streaming ingestion, enriched threat intelligence, and additional machine learning models such as autoencoders or supervised classifiers.

12 Conclusion

This project demonstrates how machine learning can enhance the capabilities of a traditional Security Operations Center. By integrating Suricata IDS, a feature extraction pipeline, an Isolation Forest anomaly detection model, and Splunk SIEM visualization, the proposed system successfully identified suspicious network behavior and converted it into useful SOC insights.

The experimental results showed that the AI-assisted workflow processed 5000 events and identified 250 anomalies, which were then visualized and operationalized through a dashboard and alerting mechanism. This confirms the feasibility of combining IDS telemetry with machine learning and SIEM visualization in a practical cybersecurity lab environment.

Future work may focus on real-time streaming pipelines, more advanced feature engineering, comparison with additional machine learning models, and deployment in larger enterprise-like environments.

13 References

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